

PIPE-Cypher: Automatic Enterprise Benchmark Generation for Text-to-Cypher Systems

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Abstract

Enterprises increasingly use property graphs and Cypher to analyze fraud, risk, identity, customer, and operational data. Public Text2Cypher benchmarks rarely reflect private schemas, terminology, access patterns, or difficulty distributions. We present PIPE-Cypher, a local-model pipeline for generating organization-specific natural-language-to-Cypher benchmarks. PIPE-Cypher combines schema introspection, reverse-query grounding, constrained Cypher generation, deterministic read-only and schema validation, execution feedback, repair, diversity controls, and LLM-judge review. The system is designed for industry settings where benchmark generation must avoid paid APIs, preserve data governance, and remain repeatable as graphs evolve.

1 Introduction

Property graphs encode enterprise relationships directly: account transfers, identity entitlements, access paths, customer interactions, and fraud rings. Cypher is a common query language for these graphs. As LLMs become natural-language interfaces for graph analytics, organizations need reliable benchmarks for natural-language-to-Cypher systems on their own schemas.

Static public benchmarks are insufficient for this setting. They cannot contain private labels, relationship types, categorical values, or domain-specific wording, and they do not track schema evolution. PIPE-Cypher addresses this gap by generating private, repeatable, execution-grounded NL-Cypher benchmarks.

2 Related Work

Recent Text2Cypher resources, including Mind the Query (Chauhan et al., 2025), SyntheT2C (Zhong et al., 2025), Auto-Cypher (Tiwari et al., 2025), and Text2Cypher (Ozsoy et al., 2025), show that

high-quality Cypher data requires more than asking an LLM for query pairs. These works motivate schema grounding, execution checks, synthetic generation, verification, and complexity-aware evaluation. PIPE-Cypher differs by targeting private enterprise benchmark generation as the primary artifact: organizations bring their own graph, run local models, enforce Cypher constraints, and receive an executable benchmark with diversity and judge metadata.

Text-to-SQL benchmarks such as Spider 2.0 (Lei et al., 2024) and BIRD (Li et al., 2023) motivate realistic database tasks, execution-based evaluation, and enterprise workflow complexity. CIKM AutoQuery (Zheng et al., 2024) motivates strategy-level workload generation analysis and separating workload quality from model quality. LDBC FinBench (Qi et al., 2023) and SNB (Püroja et al., 2023) provide graph workloads suitable for industry-oriented evaluation.

For generated benchmark quality, PIPE-Cypher combines text-generation diversity diagnostics with text-to-query structure metrics. Distinct-n (Li et al., 2016) and self-BLEU-style redundancy checks (Zhu et al., 2018) capture lexical repetition, while schema coverage, query-signature diversity, structural feature rates, and normalized entropy over graph/category/difficulty cells capture properties that matter for executable Cypher benchmarks.

3 Method

PIPE-Cypher has five stages: schema profiling, template generation, reverse Cypher grounding, constrained Cypher generation and repair, deterministic validation and execution, and LLM-judge review. Deterministic checks enforce read-only safety, syntax shape, schema-visible labels and properties, explicit relationship types, observed relationship directions, schema-provided categorical values, and non-empty execution where required. A

lightweight Cypher analyzer extracts return aliases, variables, labels, relationship observations, risky constructs, and rewrite skip reasons so normalization is auditable rather than a silent string edit. The direction gate interprets both outgoing and incoming Cypher arrow syntax before schema checking, and rejects undirected relationship patterns so directionality is an executable constraint rather than only a prompt instruction.

4 Implementation

The implementation is a Python package with Neo4j as the experimental backend. The paper frames the contribution around Cypher and property graphs rather than a single vendor. Local models are served through a vLLM/OpenAI-compatible endpoint on ds-serv6, using Qwen3.5 models for generation and judging.

The intended full-quality model is Qwen3.5-35B-A3B, but our June 1, 2026 live evidence uses the cached Qwen3.5-9B fallback. The 35B-A3B weights were staged on ds-serv6 under root-backed storage. A serving-capacity check estimated that the staged weights require four A5000 GPUs under our vLLM budget, while only one GPU was safely free in the live snapshot, so the reported generation and downstream results use the 9B fallback.

The built-in FinBench profile is grounded in the public datagen snapshot export and includes typed node and relationship properties. Live schema inspection also records low-cardinality string values as categorical constraints for prompting and validation. The generated Neo4j import script uses uniqueness constraints for nodes and creates, rather than merges, transaction relationships so repeated account-to-account events are preserved. The SNB reference profile is grounded in the official Neo4j/Cypher headers and read-query files.

For generation, PIPE-Cypher uses seeded workload templates with deterministic reverse-binding queries and a mixed mode that prepends proven workload seeds to LLM-proposed templates. Every run records accepted and rejected candidates for yield analysis. Retrieved few-shot examples replace graph-specific values with typed placeholders, preserving query structure while reducing tenant-value leakage and memorized entity reuse. A dependency-light value grounder adds typed prompt annotations for categorical values and reverse-bound entities, covering punctuation variants, possessives, plurals, synonyms, name par-

tials, and small typos.

For LLM-judge review, the implementation slices schema context to the labels, relationship types, and properties used by the candidate query. This preserves validation against the full schema while keeping local 9B fallback-model prompts within context limits on larger schemas.

The deterministic Cypher layer borrows from production lessons in the BalkanID Cypher system: schema-only prompting, exact matching, relationship direction discipline, `RETURN DISTINCT`, reserved variable rejection, categorical values, required contextual return columns, fuzzy value annotations, placeholderized retrieval examples, and parser-aware rewrite boundaries. PIPE-Cypher records parser-style structure features and skips rewrites for risky constructs such as `UNION`, `CALL`, `UNWIND`, `WHERE EXISTS`, multiple `WHERE` clauses, or reserved variables.

We also estimate seed-template capacity before full generation. This check caught and fixed a scale blocker: reverse-binding execution was hard-capped to 10 rows, and no-slot negation/ranking seeds could not support full category targets. PIPE-Cypher now uses the configured binding limit and includes slotted negation and ranking seeds for both graphs.

5 Experiments

The generated benchmark contains 3,000 accepted examples: 2,000 from LDBC FinBench and 1,000 from LDBC SNB. Categories include simple retrieval, complex retrieval, simple aggregation, complex aggregation, boolean existence, negation/difference, path/temporal transaction, and ranking/top-k.

6 Results

The full live run produced 3,000 accepted examples from 4,777 candidates using local Qwen3.5-9B for generation and judging. Category-specific recovery top-ups filled the only under-target categories from the initial sequential run. Every exported example passed read-only, syntax, schema, execution, non-empty result, and judge gates.

The accepted full-run records were exported into a benchmark package with stable example identifiers, train/dev/test splits, result samples, gate metadata, aggregate statistics, and a manifest hash for artifact tracking. The export contains 3,000 accepted examples: 2,000 FinBench, 1,000 SNB, and

Gate	Evidence checked	Failure mode caught
Read-only safety	blocked Cypher write/admin tokens	destructive or operational queries
Schema validity	labels, relationship types, properties, categorical values	hallucinated graph vocabulary or values
Direction validity	observed relationship directions	reversed or invalid traversals
Cypher analysis and normalization	structure extraction, rewrite safety, RETURN DISTINCT	unsafe rewrites or duplicate/noisy rows
Question constraints	quoted values use exact literals	fuzzy matching of exact user values
Execution	query runs and returns rows	syntactic validity without answerability
LLM judge	ambiguity, semantic alignment, schema use	executable but semantically weak examples

Table 1: PIPE-Cypher candidate acceptance gates.

Graph	Target/category	Binding limit	Min seed capacity	All categories meet target
FinBench	250	300	300	Yes
SNB	125	200	200	Yes

Table 2: Built-in seed-template capacity after removing the reverse-binding 10-row cap and adding slotted negation/ranking seeds. Capacity is a theoretical scale-readiness check; final examples still must pass validation, execution, diversity, and judge gates.

Setting	Value
Accepted examples	3,000
Primary graph	LDBC FinBench, 2,000 examples
Secondary graph	LDBC SNB, 1,000 examples
Categories	8 balanced categories, 375 each
Generation/judge model	Local Qwen3.5-9B fallback
Execution backend	Neo4j Community, two live databases
Judge audit packet	80 sampled accepted/rejected pairs

Table 3: Full live experimental setup used for the generated benchmark artifact.

375 accepted examples in every planned category.

We report diversity as a diagnostic, not only as a design target. The full export has perfect category balance and near-perfect difficulty balance, uses 1,115 unique grounded entity values, and exactly quotes grounded values in 82.6% of examples with entity bindings. The Qwen3.5-9B fallback run still has low query-signature diversity because seeded graph-grounded templates are doing substantial work. This is useful evidence for industry deployment: the artifact is balanced and executable, while the appendix makes template and value concentration visible rather than hiding it.

The full-run failure taxonomy is concentrated in diversity/duplicate controls during recovery and empty execution results; only 2/4,777 candidates were schema-invalid after deterministic Cypher governance.

For judge calibration, the released artifacts include an 80-row audit packet sampled from accepted and rejected full-run candidates, with a 40/40 judge accept/reject split and coverage over

Graph	Candidates	Accepted	Acceptance	Categories at target
FinBench	3,404	2,000	0.588	8/8
SNB	1,373	1,000	0.728	8/8
Total	4,777	3,000	0.628	16/16

Table 4: Full live generation with local Qwen3.5-9B. Candidate counts include the initial sequential run and category-specific recovery top-ups.

Export artifact	Examples	FinBench	SNB	Train	Dev	Test
Live full benchmark	3,000	2,000	1,000	2,408	296	296

Table 5: Accepted live full benchmark package with stable IDs, gate metadata, result samples, statistics, and manifest hash 8bc79a53a06b291a.

both graphs and all eight categories. The calibration tooling tracks graph, category, difficulty, strategy, and judge-decision coverage, and reports agreement, precision/recall, specificity, balanced accuracy, and Cohen’s kappa once human labels are filled. The current draft treats these labels as pending human review rather than as a generation gate.

Finally, we ran a downstream Text2Cypher evaluation using local Qwen3.5-9B on the exported full test split. The model saw schema text and the natural-language question, generated Cypher, and was evaluated by live execution against the corresponding FinBench or SNB database. The result verifies that the exported artifact can drive downstream model evaluation and gives a difficulty signal: the zero-shot 9B baseline solves simple aggregation examples but struggles on more operational categories.

7 Industry Use

PIPE-Cypher is intended to be rerun when an enterprise graph changes. The generated artifact records the schema snapshot, graph profile, model identifier, validation gates, execution samples, judge scores, difficulty features, and source run for ev-

Artifact property	Value
Categories	8 balanced categories
FinBench/category	250
SNB/category	125
Difficulty split	1,521 easy / 1,479 medium
Unique labels / rel. types	7 / 9
Read/syntax/schema/exec/judge gates	3,000/3,000

Table 6: Distribution and gate summary for the exported full benchmark artifact.

Split	Examples	Parse	Schema	Exec. success	Exec. acc.	Answer F1
Live full test	296	0.959	0.905	0.622	0.189	0.189

Table 7: Downstream Text2Cypher evaluation for local Qwen3.5-9B on the exported full benchmark test split.

ery example. Long-running jobs can be monitored from JSONL records and recovered with category-specific top-up runs that reject questions accepted in earlier runs. This design supports private benchmark refreshes without exposing schemas or values to paid APIs, while still leaving audit hooks for data owners to sample accepted and rejected examples.

8 Conclusion

PIPE-Cypher provides a practical path for enterprises to create private, executable, balanced NL-to-Cypher benchmarks. The pipeline combines schema introspection, constrained generation, deterministic validation, execution feedback, diversity control, and automated judge review.

Limitations

Execution validity does not guarantee semantic correctness. LLM judges can over-accept plausible but wrong examples, so PIPE-Cypher releases a frozen post-hoc audit packet and requires complete human labels before reporting judge-human agreement. Generated artifacts may contain private schema details or sampled values, so organizations should apply normal data governance controls.

Ethics Statement

PIPE-Cypher is designed for private benchmark generation under local-model deployment constraints. Organizations using it should restrict benchmark artifacts to authorized users, review sampled values for sensitive content, and document whether judge calibration relied on human annotators.

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Metric	Value
Question Distinct-1	0.041
Question Distinct-2	0.088
Question self-BLEU-2 (sampled)	0.826
Unique query-signature ratio	0.010
Top query-signature share	0.083
Category normalized entropy	1.000
Graph-category normalized entropy	0.980
Difficulty normalized entropy	1.000
Label coverage	0.412
Relationship-type coverage	0.375
Property-name coverage	0.407
Unique grounded-value ratio	0.373
Grounded values exactly quoted	0.826
Aggregation / negation / ordering rates	0.500 / 0.125 / 0.125

Table 8: Diversity diagnostics for the full exported benchmark. Distinct-n follows text-generation usage; self-BLEU is lower when questions are less redundant; grounded-value metrics are aggregate-only and do not list raw values.

Failure bucket	Count	Share of rejected
Diversity/duplicate control	1,335	0.751
Empty result	374	0.210
Judge semantic reject	66	0.037
Schema invalid	2	0.001

Table 9: Failure taxonomy over full-run generation candidates before benchmark export. Accepted examples are excluded from the bucket shares.

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Setting	Graph	Records	Accepted	Acceptance	Categories at target
Unconstrained LLM	FinBench	0	0	0.000	0/8
Unconstrained LLM	SNB	0	0	0.000	0/8
Reverse-only	FinBench	400	400	1.000	8/8
Reverse-only	SNB	402	400	0.995	8/8
Validators+repair	FinBench	400	400	1.000	8/8
Validators+repair	SNB	402	400	0.995	8/8
No retrieval	FinBench	429	400	0.932	8/8
No retrieval	SNB	402	400	0.995	8/8
No rewrite	FinBench	423	400	0.946	8/8
No rewrite	SNB	402	400	0.995	8/8
No LLM judge	FinBench	401	400	0.998	8/8
No LLM judge	SNB	406	400	0.985	8/8
Full PIPE-Cypher	FinBench	416	400	0.962	8/8
Full PIPE-Cypher	SNB	402	400	0.995	8/8

Table 10: Live target-50 ablation evidence with local Qwen3.5-9B. Each graph run targets 50 accepted examples per category.

Setting	Graph	Read-only	Syntax	Schema	Exec.	Judge
Unconstrained LLM	FinBench	0.000	0.000	0.000	0.000	0.000
Unconstrained LLM	SNB	0.000	0.000	0.000	0.000	0.000
Reverse-only	FinBench	1.000	1.000	1.000	1.000	1.000
Reverse-only	SNB	1.000	1.000	0.995	0.995	0.995
Validators+repair	FinBench	1.000	1.000	1.000	1.000	1.000
Validators+repair	SNB	1.000	1.000	0.995	0.995	0.995
No retrieval	FinBench	1.000	1.000	1.000	1.000	0.932
No retrieval	SNB	1.000	1.000	0.995	0.995	0.995
No rewrite	FinBench	1.000	1.000	1.000	1.000	0.946
No rewrite	SNB	1.000	1.000	0.995	0.995	0.995
No LLM judge	FinBench	1.000	1.000	1.000	1.000	0.998
No LLM judge	SNB	1.000	1.000	0.995	0.995	0.985
Full PIPE-Cypher	FinBench	1.000	1.000	1.000	1.000	0.962
Full PIPE-Cypher	SNB	1.000	1.000	0.995	0.995	0.995

Table 11: Quality-gate rates for the live target-50 ablation suite. Rates are computed over all generated records in each graph/setting.

A Additional Results

B Claim–Evidence Map

This section maps the main paper claims to the strongest current evidence and to the remaining risks. This is intended as a reviewer-facing audit surface: claims with pending evidence remain marked as such rather than being folded into the main results.

- Claim 1.** PIPE-Cypher can generate a private, executable NL-to-Cypher benchmark over enterprise-style property graphs using local models.

Evidence. The full Qwen3.5-9B fallback run exported 3,000 accepted examples: 2,000 FinBench and 1,000 SNB, with 375 examples in every planned workload category and all accepted examples passing read-only, syntax, schema, execution, non-empty-result, and judge gates.

Key artifacts. benchmark export; manifest snapshot; paper table: benchmark export; paper table: full results

Status. Supported by live 9B fallback evidence; 35B target run is blocked by current GPU capacity.

Audit packet property	Value
Rows	80
Judge accept / reject	40 / 40
FinBench / SNB rows	48 / 32
Easy / medium rows	26 / 54
Labeled rows	0
Calibration status	unlabeled

Table 12: Post-hoc judge calibration packet coverage. Human labels are pending until the packet is complete and are not used as a generation gate.

Metric	Point	95% CI	SE	N
Parse valid	0.959	[0.936, 0.980]	0.012	296
Schema valid	0.905	[0.872, 0.939]	0.017	296
Execution success	0.622	[0.564, 0.676]	0.028	296
Execution accuracy	0.189	[0.145, 0.233]	0.022	296
Answer F1	0.189	[0.149, 0.236]	0.023	296

Table 13: Bootstrap uncertainty for downstream Text2Cypher evaluation on the full exported test split. Intervals use row-level resampling with a fixed seed and are intended for appendix reporting.

Risk. Generation quality may change with the target 35B model or with private enterprise schemas beyond FinBench/SNB.

- Claim 2.** Cypher-specific governance makes generated benchmark candidates safer and more auditable than prompt-only generation.

Evidence. The pipeline validates read-only safety, schema-visible labels/properties/relationships, observed relationship direction, categorical values, contextual return columns, parser-style structure, execution success, and LLM-judge review. In the full run, only 2 of 4,777 candidates were schema-invalid after deterministic governance, and all 3,000 exported examples passed the deterministic and judge gates.

Key artifacts. code: validator.py; code: cypher_parser.py; code: question_constraints.py; snapshot: failure taxonomy; +1 more

Status. Supported by code, tests, and full-run failure taxonomy.

Risk. Semantic correctness still depends on execution evidence and judge review; human audit labels are pending.

- Claim 3.** Reverse grounding and structured workload seeds are designed to improve reliability under local-model constraints.

Evidence. The target-50 FinBench/SNB ablation suite completed 14/14 graph/variant cells, reached every category target for all non-empty/non-unconstrained runs, and passed the paper-readiness audit with logs, model IDs, code revision, run summaries, and gate-rate availability. Appendix-only tables plus yield and gate-rate figures report the audited target-50 evidence while the target-100 follow-up and seeded target-50 repeat continue for stronger final reliability and variance claims.

Key artifacts. script: run live ablation suite; script: monitor remote ablation suite; script: monitor remote ablation queue; script: collect remote ablation suite; +9 more

Status. Supported by audited target-50 appendix evidence; target-100 and seeded target-50 repeat runs are still pending for stronger final reliability claims.

Risk. A single target-50 suite is the minimum publishable ablation scale, not the strongest possible evidence. If target-100 or repeated-seed runs complete, final claims should use those sensitivity results instead of relying only on target-50.

- Claim 4.** The benchmark controls category and difficulty balance while exposing residual diversity limits.

Evidence. The full export has perfect category balance, planned 2:1 FinBench/SNB graph mix, and a near-even easy/medium split. Diversity diagnostics report Distinct-n, sampled self-BLEU-2, query-signature diversity, normalized entropy, schema coverage, structural feature rates, and aggregate value-grounding diagnostics. The full export uses 1,115 unique grounded entity values and exactly quotes grounded values in 82.6% of examples with entity bindings, making template and value concentration visible instead of hidden.

Key artifacts. snapshot: diversity report; paper table: diversity; paper figure: diversity diagnostics; paper figure: full export distribution

Status. Supported by full-export statistics and diversity diagnostics.

436	<i>Risk.</i> Low query-signature diversity in the 9B	agreement remains unreported until all audit	485
437	fallback run remains a limitation and ablation	rows are labeled.	486
438	target.		
439	5. Claim 5. The generated benchmark is useful	<i>Risk.</i> Judge-human agreement is not yet	487
440	for downstream Text2Cypher evaluation.	known because human labels are still blank.	488
441	<i>Evidence.</i> A zero-shot local Qwen3.5-9B	7. Claim 7. The intended 35B local genera-	489
442	downstream evaluation over the 296-example	tion/judge path is staged but not currently	490
443	full test split reports parse validity, schema	runnable under observed shared-GPU con-	491
444	validity, execution success, execution accu-	straints.	492
445	racy, answer F1, per-category breakdowns,	<i>Evidence.</i> Qwen3.5-35B-A3B is staged on ds-	493
446	and fixed-seed bootstrap uncertainty inter-	serv6, but the capacity checker reports 68,573	494
447	vals. The model reaches 0.189 execution accu-	MiB of weights, four required A5000 GPUs	495
448	racy with a 95% bootstrap interval of [0.145,	under the conservative vLLM budget, and	496
449	0.233] and 0.622 execution success with in-	only one safe GPU in the latest snapshot.	497
450	terval [0.564, 0.676], showing the benchmark	<i>Key artifacts.</i> script: check vllm capacity;	498
451	can discriminate easy aggregation from harder	script: render vllm capacity snapshot; re-	499
452	operational categories.	search note: model availability; research note:	500
453	<i>Key artifacts.</i> downstream summary; snap-	qwen35b capacity snapshot 20260601; +1	501
454	shot: downstream uncertainty; research note:	more	502
455	downstream evaluation; paper table: down-	<i>Status.</i> Documented blocker; current paper	503
456	stream evaluation; +3 more	results are explicitly reported as Qwen3.5-9B	504
457	<i>Status.</i> Supported by live downstream evalua-	fallback evidence.	505
458	tion against FinBench/SNB databases, with	<i>Risk.</i> A later 35B run may change yield, diver-	506
459	bootstrap uncertainty computed from row-	sity, judge behavior, and downstream conclu-	507
460	level outputs.	sions.	508
461	<i>Risk.</i> Only one downstream model has been		
462	evaluated so far.	C Prompt Contracts	509
463	6. Claim 6. Automated LLM-judge review can	PIPE-Cypher treats prompts as versioned imple-	510
464	replace human-in-the-loop generation gates	mentation artifacts. The list summarizes the prompt	511
465	while still exposing judge risk.	contracts used for generation, repair, judging, and	512
466	<i>Evidence.</i> The pipeline uses deterministic val-	downstream evaluation; hashes fingerprint the full	513
467	idation plus LLM-judge review as the gener-	prompt constants in the codebase.	514
468	ation gate. A post-hoc 80-row judge audit	1. Template generation. Stage: Workload pro-	515
469	packet preserves 40 judge accepts, 40 judge	posal. SHA-256: 10b25b.	516
470	rejects, both graphs, and all eight categories.	<i>Contract.</i> Schema-only labels, relationships,	517
471	Deterministic sidecar generation now creates	properties, and categorical values; Realistic	518
472	two independent annotator sheets plus an ad-	enterprise analyst wording; At most two typed	519
473	judication template, with a tracked hash mani-	slots and JSON-only output	520
474	fest that does not commit raw value-bearing		
475	sheets. The analyzer reports agreement, preci-	2. Reverse binding. Stage: Graph grounding.	521
476	sion/recall, specificity, balanced accuracy, and	SHA-256: 48e949.	522
477	Cohen’s kappa only after the packet is com-	<i>Contract.</i>	
478	pletely human-labeled; partial labels are not	Read-only	523
479	treated as paper-ready calibration.	MATCH/WHERE/RETURN DIS-	524
480	<i>Key artifacts.</i> code: judge.py; code: calibra-	TINCT/LIMIT only; Slot variables named	525
481	tion.py; code: judge_audit_packet.py; script:	exactly as requested; Forward relationship	526
482	create judge annotation sheets; +4 more	directions from the schema	527
483	<i>Status.</i> Pipeline gate is implemented; audit	3. Cypher generation. Stage: Candidate query.	528
484	packet coverage is supported; judge-human	SHA-256: 61c557.	529

530	<i>Contract.</i> Only schema-visible constructs and	2. FinBench / Complex Aggregation / medium.	577
531	observed directions; RETURN DISTINCT for	<i>Question:</i> What is the total transferred amount	578
532	set returns and exact equality for quoted val-	from accounts owned by person 'Gwar'?	579
533	ues; Context columns, categorical hints, place-		
534	holderized retrieval, and no writes	<pre>MATCH (p:Person {personName: 'Gwar'}) -[:OWN_ACCOUNT]-> (src:Account)-[t:TRANSFER_TO]-> (:Account) RETURN DISTINCT SUM(t.amount) AS TotalTransferredAmount</pre>	580 581 582 583 584 585 586
535	4. Repair. Stage: Validation feedback. SHA-	<i>Structure:</i> join_heavy, aggregation.	587
536	256: 56c419.	<i>Relationships:</i> OWN_ACCOUNT, TRANS-	588
537	<i>Contract.</i> Preserve question intent while fix-	FER_TO.	589
538	ing validation or execution issues; Keep query	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.	590
539	read-only and schema-grounded; Return only	<i>Result sample:</i> {TotalTransferredAmount:	591
540	corrected Cypher	14972318.41}; observed rows: 1	592
541	5. LLM judge. Stage: Quality gate. SHA-256:	3. FinBench / Complex Retrieval / easy. Ques-	593
542	073938.	<i>tion:</i> Which accounts received transfers from	594
543	<i>Contract.</i> Inputs include question, Cypher,	accounts owned by person 'Zof'?	595
544	schema slice, execution rows, and validation		
545	summary; Strict JSON scores for ambiguity,	<pre>MATCH (p:Person {personName: 'Zof'}) -[:OWN_ACCOUNT]-> (src:Account) -[:TRANSFER_TO]-> (dst:Account) RETURN DISTINCT dst.accountId AS AccountId, dst.accountType AS AccountType, dst.isBlocked AS IsBlocked LIMIT 300</pre>	596 597 598 599 600 601 602 603 604
546	semantic alignment, schema use, and diffi-	<i>Structure:</i> join_heavy, bounded_result.	605
547	culty; Pass only useful, unambiguous enter-	<i>Relationships:</i> OWN_ACCOUNT, TRANS-	606
548	prise benchmark examples	FER_TO.	607
549	6. Downstream Text2Cypher. Stage: Model	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.	608
550	evaluation. SHA-256: 4c07ff.	<i>Result sample:</i> {AccountId:	609
551	<i>Contract.</i> Read-only Cypher only; Schema-	4687402787162554854, AccountType:	610
552	visible constructs and exact direction preser-	credit card, IsBlocked: False}; observed rows:	611
553	vation; RETURN DISTINCT, count/ranking	3	612
554	rules, and no explanations	4. FinBench / Negation Difference / medium.	613
555	D Representative Accepted Examples	<i>Question:</i> Which accounts owned by person	614
556	The examples below are selected in stable identifier	'Kant' have not sent any transfers?	615
557	order from the tracked full-export snapshot, one per		
558	graph/category cell when available. They show the	<pre>MATCH (p:Person {personName: 'Kant'}) -[:OWN_ACCOUNT]-> (a:Account) WHERE NOT (a) -[:TRANSFER_TO]-> (:Account) RETURN DISTINCT a.accountId AS AccountId, a.accountType AS AccountType, a.isBlocked AS IsBlocked LIMIT 300</pre>	616 617 618 619 620 621 622 623 624 625
559	natural-language question, accepted Cypher, struc-	<i>Structure:</i> join_heavy, negation,	626
560	tural tags, gate status, and a bounded execution-	bounded_result.	627
561	result sample.	<i>Relationships:</i> OWN_ACCOUNT, TRANS-	628
562	1. FinBench / Boolean Existence / easy. Ques-	FER_TO.	629
563	<i>tion:</i> Does account '187743809466009406'	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.	630
564	have any outgoing transfer?	<i>Result sample:</i> {HasOutgoingTransfer: True};	631
565	<pre>MATCH (src:Account {accountId: '187743809466009406'}) -[:TRANSFER_TO]-> (:Account) RETURN DISTINCT COUNT(DISTINCT src) > 0 AS HasOutgoingTransfer</pre>	4739757132830738341, AccountType:	632
566			
567			
568			
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571			
572	<i>Structure:</i> single_hop, aggregation.		
573	<i>Relationships:</i> TRANSFER_TO.		
574	<i>Gates:</i> RO/Syn/Schema/Exec/Judge.		
575	<i>Result sample:</i> {HasOutgoingTransfer: True};		
576	observed rows: 1		

633	merchant account, IsBlocked: False};	Structure: single_hop, aggregation.	691
634	observed rows: 1	Relationships: OWN_ACCOUNT.	692
635	5. FinBench / Path Temporal / medium. Question: Which accounts can receive money	Gates: RO/Syn/Schema/Exec/Judge.	693
636	within two transfer hops from accounts owned	Result sample: {AccountCount: 1}; observed	694
637	by person 'Sossamon'?	rows: 1	695
638			
639	<pre>MATCH (p:Person {personName: 640 'Sossamon'}) -[:OWN_ACCOUNT]-> 641 (src:Account) -[:TRANSFER_TO*1..2]-> 642 (dst:Account) 643 RETURN DISTINCT dst.accountId AS 644 AccountId, dst.accountType AS 645 AccountType, dst.isBlocked AS 646 IsBlocked 647 LIMIT 300</pre>	8. FinBench / Simple Retrieval / easy. Question: Which accounts are owned by person	696
648	Structure: join_heavy, path, bounded_result.	'Barry'?	697
649	Relationships: OWN_ACCOUNT, TRANS-	<pre>MATCH (p:Person {personName: 699 'Barry'}) 700 -[:OWN_ACCOUNT]-> (a:Account) 701 RETURN DISTINCT a.accountId AS 702 AccountId, a.accountType AS 703 AccountType, 704 a.isBlocked AS IsBlocked 705 LIMIT 300 706</pre>	698
650	FER_TO.	Structure: single_hop, bounded_result.	707
651	Gates: RO/Syn/Schema/Exec/Judge.	Relationships: OWN_ACCOUNT.	708
652	Result sample: {AccountId:	Gates: RO/Syn/Schema/Exec/Judge.	709
653	4687402787162554854, AccountType:	Result sample: {AccountId:	710
654	credit card, IsBlocked: False}; observed rows:	4732438783436260772, AccountType:	711
655	4	internet account, IsBlocked: False}; observed	712
656	6. FinBench / Ranking Topk / medium. Question: For accounts owned by person 'Barry',	rows: 1	713
657	which account sent the highest total transfer	9. SNB / Boolean Existence / medium. Question: Does person with id 6597069766828	714
658	amount?	like any post?	715
659		<pre>MATCH (p:Person {id: 717 6597069766828}) 718 OPTIONAL MATCH (p) -[:LIKES]-> 719 (post:Post) 720 RETURN DISTINCT COUNT(post) > 0 AS 721 LikesAnyPost 722</pre>	716
660	<pre>MATCH (p:Person {personName: 661 'Barry'}) 662 -[:OWN_ACCOUNT]-> 663 (src:Account) -[t:TRANSFER_TO]-> 664 (:Account) 665 WITH src, SUM(t.amount) AS 666 totalAmount 667 RETURN DISTINCT src.accountId AS 668 AccountId, src.accountType AS 669 AccountType, src.isBlocked AS 670 IsBlocked, 671 totalAmount 672 ORDER BY totalAmount DESC 673 LIMIT 1</pre>	Structure: single_hop, aggregation, optional.	723
674	Structure: join_heavy, aggregation, or-	Relationships: LIKES.	724
675	der_rank, bounded_result.	Gates: RO/Syn/Schema/Exec/Judge.	725
676	Relationships: OWN_ACCOUNT, TRANS-	Result sample: {LikesAnyPost: True}; ob-	726
677	FER_TO.	served rows: 1	727
678	Gates: RO/Syn/Schema/Exec/Judge.	10. SNB / Complex Aggregation / medium.	728
679	Result sample: {AccountId:	Question: How many distinct posts	729
680	4732438783436260772, AccountType:	are in forums joined by person with id	730
681	internet account, IsBlocked: False}; observed	6597069766845?	731
682	rows: 1	<pre>MATCH (forum:Forum) -[:HAS_MEMBER]-> 732 (p:Person {id: 6597069766845}) 733 MATCH (forum) -[:CONTAINER_OF]-> 734 (post:Post) 735 RETURN DISTINCT COUNT(DISTINCT post) 736 AS 737 JoinedForumPostCount 738</pre>	732
683	7. FinBench / Simple Aggregation / easy. Question: How many accounts are owned by per-	Structure: join_heavy, aggregation.	739
684	son 'Kaewsuktae'?	Relationships: CONTAINER_OF,	740
685		HAS_MEMBER.	741
686	<pre>MATCH (p:Person {personName: 687 'Kaewsuktae'}) -[:OWN_ACCOUNT]-> 688 (a:Account) 689 RETURN DISTINCT COUNT(DISTINCT a) AS 690 AccountCount</pre>	Gates: RO/Syn/Schema/Exec/Judge.	742

743	<i>Result sample:</i> {JoinedForumPostCount: 1};	14. SNB / Ranking Topk / medium. Question:	796
744	observed rows: 1	Among forums tagged 'James_K._Polk',	797
745		which forum has the most members?	798
746	11. SNB / Complex Retrieval / easy. Question:		799
747	Which people are members of forums contain-		800
748	ing posts tagged 'Manuel_Noriega'?		801
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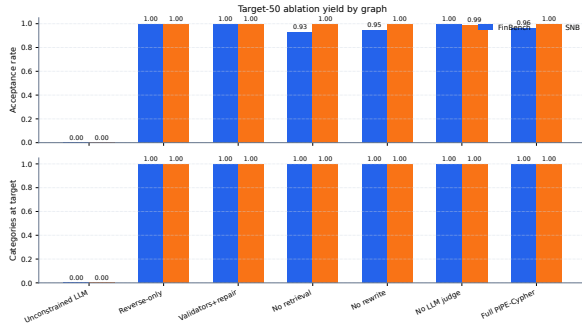


Figure 1: Audited target-50 ablation suite over FinBench and SNB. The suite reaches all graph/variant/category targets and is included as appendix evidence while the larger target-100 and seeded repeat suites continue running for stronger final reliability evidence.

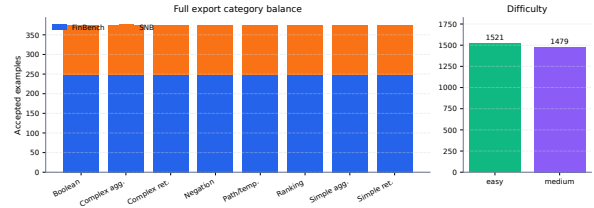


Figure 4: Full 3,000-example export distribution. The benchmark preserves the planned 2:1 FinBench/SNB graph mix while balancing all eight Cypher workload categories and maintaining a near-even easy/medium difficulty split.

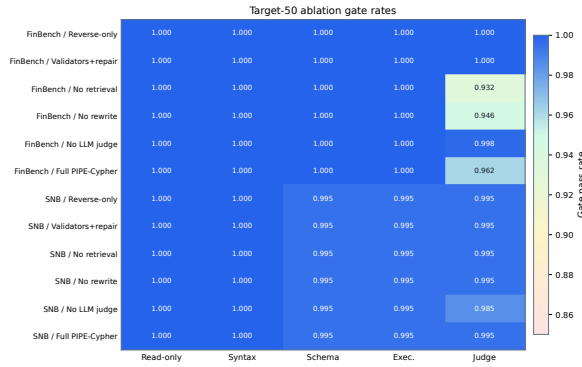


Figure 2: Gate-rate heatmap for the audited target-50 ablation suite. Deterministic read-only and syntax gates are saturated, while schema, execution, and judge rates expose the remaining quality differences across graph and pipeline variants.

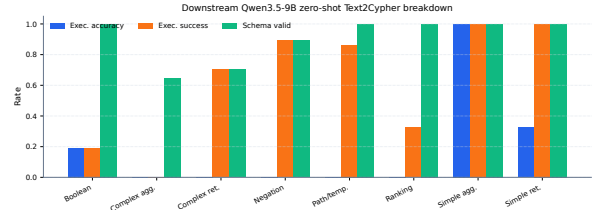


Figure 5: Downstream zero-shot Qwen3.5-9B Text2Cypher evaluation by workload category on the full test split. The breakdown separates final execution accuracy from parse/schema/execution success signals, making difficult operational categories visible.

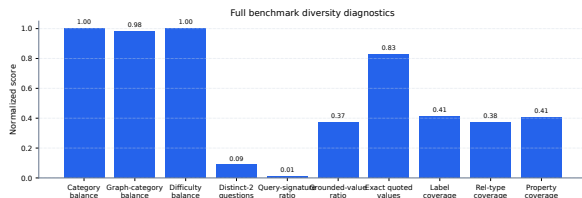


Figure 3: Diversity diagnostics for the full 3,000-example export. Category, graph-category, and difficulty balance are strong by construction; schema coverage and query-signature diversity expose remaining concentration from the fallback seeded-template run.

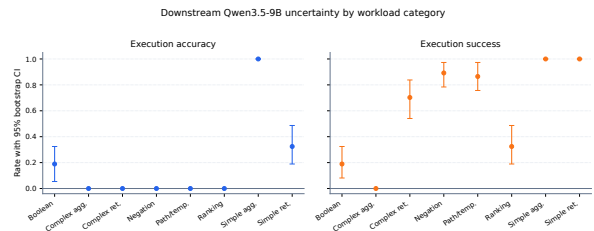


Figure 6: Bootstrap uncertainty for downstream Qwen3.5-9B Text2Cypher evaluation by workload category. Error bars show 95% row-level bootstrap intervals over the full exported test split.